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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 25th January Morning Shift

JEE Main Question Paper with Answer Key

25 Jan 2023	90	18	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	23 questions
Physics	26 questions
Out of syllabus	18 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

Inert gases have positive electron gain enthalpy. Its correct order is :

(A) He < Ne < Kr < Xe

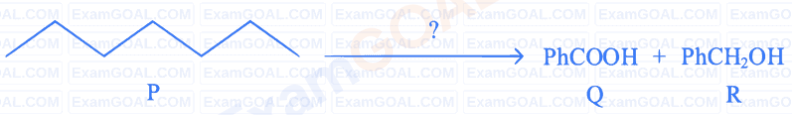
(B) He < Xe < Kr < Ne

(C) Xe < Kr < Ne < He

(D) He < Kr < Xe < Ne

Open solution

Q2. MCQ (Single Correct Answer)



Open solution

The correct sequence of reagents for the preparation of Q and R is :

(A) $(i)Cr_2O_3, 770K, 20\ atm; (ii)CrO_2Cl_2, H_3O^+; (iii)NaOH; (iv)H_3O^+$

(B) $(i)CrO_2Cl_2, H_3O^+; (ii)Cr_2O_3, 770K, 20\ atm; (iii)NaOH; (iv)H_3O^+$

(C) $(i)KMnO_4, OH^-; (ii)Mo_2O_3, \Delta; (iii)NaOH; (iv)H_3O^+$

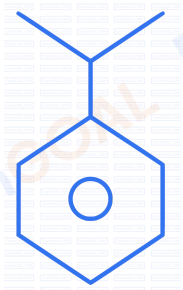
(D) $(i)Mo_2O_3, \Delta; (ii)CrO_2Cl_2, H_3O^+; (iii)NaOH; (iv)H_3O^+$

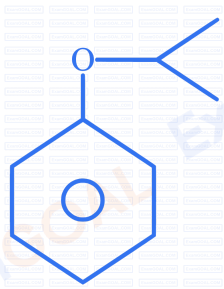
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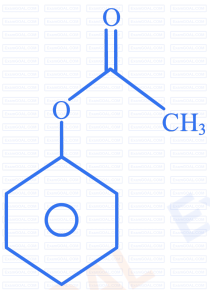
Q3. MCQ (Single Correct Answer)

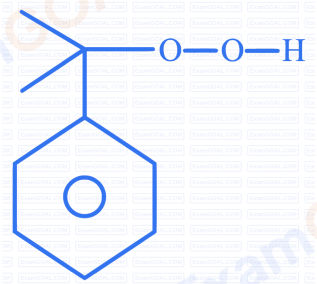
Open solution

In the cumene to phenol preparation in presence of air, the intermediate is

(A) 

(B) 

(C) 

(D) 

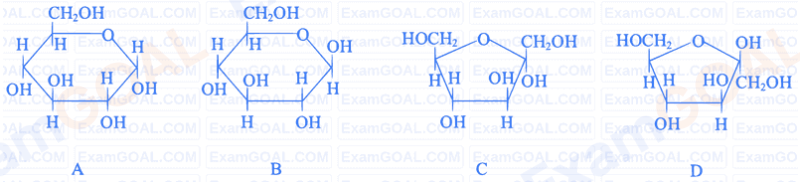
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Q4. MCQ (Single Correct Answer)

Open solution

Match items of Row I with those of Row II.

Row I :



Row II :

(i) α-D-(−)-Fructofuranose

(ii) β-D-(−)-Fructofuranose

(iii) α-D-(−)-Glucopyranose

(iv) β-D-(−)-Glucopyranose

Correct match is

(A) A → iii, B → iv, C → i, D → ii

(B) A → iv, B → iii, C → i, D → ii

(C) A → i, B → ii, C → iii, D → iv

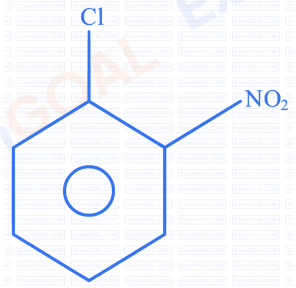
(D) A → iii, B → iv, C → ii, D → i

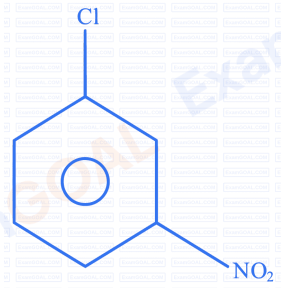
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Q5. MCQ (Single Correct Answer)

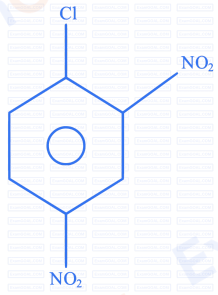
Open solution

The compound which will have the lowest rate towards nucleophilic aromatic substitution on treatment with OH^- is

(A) 

(B) 

(C) 

(D) 

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Q6. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List I Elements		List II Colour imparted to the flame
A.	K	I.	Brick Red
B.	Ca	II.	Violet
C.	Sr	III.	Apple Green
D.	Ba	IV.	Crimson Red

Choose the correct answer from the options given below :

(A) A-II, B-IV, C-I, D-III

(B) A-II, B-I, C-IV, D-III

(C) A-II, B-I, C-III, D-IV

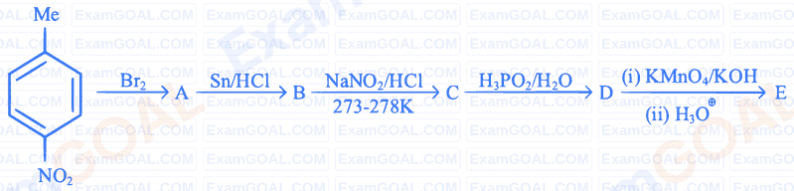
(D) A-IV, B-III, C-II, D-I

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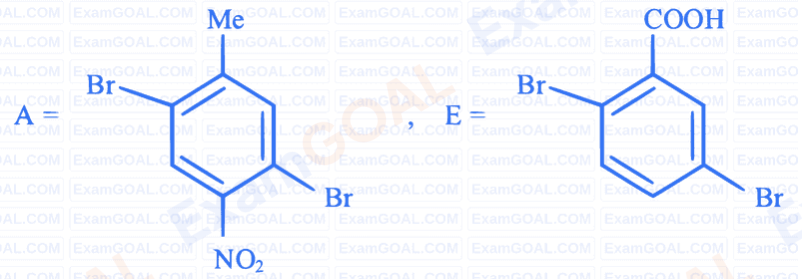
Q7. MCQ (Single Correct Answer)

Open solution

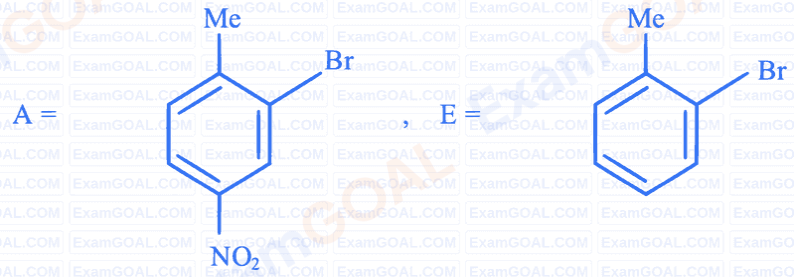
Identify the product formed (A and E)



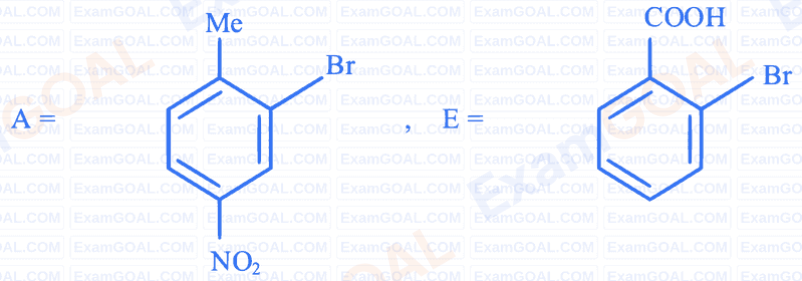
(A)



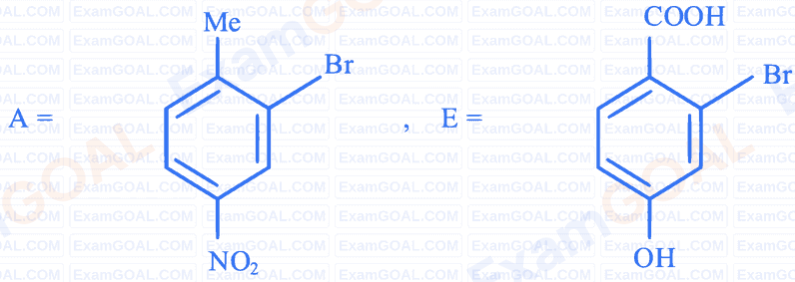
(B)



(C)



(D)



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Q8. MCQ (Single Correct Answer)

'25 volume' hydrogen peroxide means

(A) 1 L marketed solution contains 75 g of H_2O_2.

(B) 1 L marketed solution contains 250 g of H_2O_2.

(C) 1 L marketed solution contains 25 g of H_2O_2.

(D) 100 mL marketed solution contains 25 g of H_2O_2.

Open solution

Q9. MCQ (Single Correct Answer)

The radius of the 2nd orbit of Li^{2+} is x. The expected radius of the 3rd orbit of Be^{3+} is

(A) $\frac{16}{27}x$ (B) $\frac{4}{9}x$ (C) $\frac{9}{4}x$ (D) $\frac{27}{16}x$

Open solution

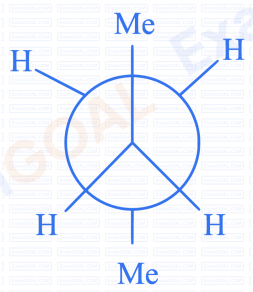
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Q10. MCQ (Single Correct Answer)

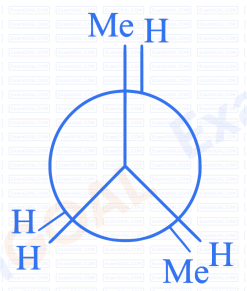
Open solution

Which of the following conformations will be the most stable?

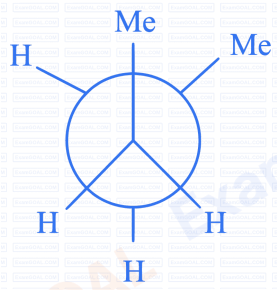
(A)



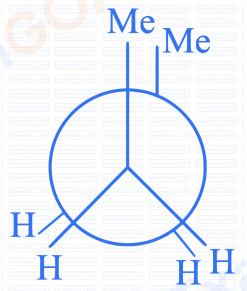
(B)



(C)



(D)



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Q11. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Acetal / Ketal is stable in basic medium.

Reason R : The high leaving tendency of alkoxide ion gives the stability to acetal/ketal in basic medium.

In the light of the above statements, choose the correct answer from the options given below :

(A) A is false but R is true.

(B) Both A and R are true but R is NOT the correct explanation of A

(C) Both A and R are true and R is the correct explanation of A

(D) A is true but R is false.

Q12. MCQ (Single Correct Answer)

Open solution

The correct order in aqueous medium of basic strength in case of methyl substituted amines is :

(A) $NH_3 > Me_3N > MeNH_2 > Me_2NH$

(B) $Me_3N > Me_2NH > MeNH_2 > NH_3$

(C) $Me_2NH > MeNH_2 > Me_3N > NH_3$

(D) $Me_2NH > Me_3N > MeNH_2 > NH_3$

Q13. MCQ (Single Correct Answer)

Open solution

Match the List-I with List-II :

	List-I Cations		**List-II Group reagents**
A.	Pb^{2+}, Cu^{2+}	i)	H_2S gas in presence of dilute HCl
B.	Al^{3+}, Fe^{3+}	ii)	$(NH_4)_2CO_3$ in presence of NH_4OH
C.	Co^{2+}, Ni^{2+}	iii)	NH_4OH in presence of NH_4Cl
D.	Ba^{2+}, Ca^{2+}	iv)	H_2S in presence of NH_4OH

Correct match is -

(A) A $\rightarrow$ i, B $\rightarrow$ iii, C $\rightarrow$ iv, D $\rightarrow$ ii

(B) A $\rightarrow$ i, B $\rightarrow$ iii, C $\rightarrow$ ii, D $\rightarrow$ iv

(C) A $\rightarrow$ iii, B $\rightarrow$ i, C $\rightarrow$ iv, D $\rightarrow$ ii

(D) A $\rightarrow$ iv, B $\rightarrow$ ii, C $\rightarrow$ iii, D $\rightarrow$ i

Q14. Numerical

Open solution

The density of a monobasic strong acid (Molar mass 24.2 g/mol) is 1.21 kg/L. The volume of its solution required for the complete neutralization of 25 mL of 0.24 M NaOH is _____ $\times 10^{-2}$ mL (Nearest integer)

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Q15. Numerical

A litre of buffer solution contains 0.1 mole of each of NH_3 and NH_4Cl. On the addition of 0.02 mole of HCl by dissolving gaseous HCl, the pH of the solution is found to be _____ $\times 10^{-3}$ (Nearest integer)

[Given : $pK_b(NH_3) = 4.745$

$\log 2 = 0.301$

$\log 3 = 0.477$

$T = 298\ K$]

Open solution

Q16. Numerical

Consider the cell

$Pt(s)|H_2(g)\ (1\ atm)|H^+\ (aq, [H^+] = 1)||Fe^{3+}(aq), Fe^{2+}(aq)|Pt(s)$

Given $E^\circ_{Fe^{3+}/Fe^{2+}} = 0.771\ V$ and $E^\circ_{H^+/1/2\ H_2} = 0\ V,\ T = 298\ K$

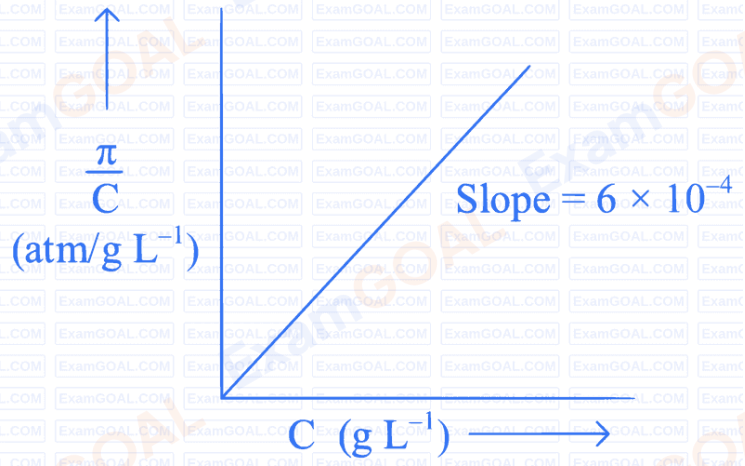
If the potential of the cell is 0.712 V, the ratio of concentration of Fe^{2+} to Fe^{3+} is _____ (Nearest integer)

Open solution

Q17. Numerical

The osmotic pressure of solutions of PVC in cyclohexanone at 300 K are plotted on the graph.

The molar mass of PVC is _____ g mol^{-1} (Nearest integer)



(Given : R = 0.083 L atm K^{-1} mol^{-1})

Open solution

Q18. Numerical

In sulphur estimation, 0.471 g of an organic compound gave 1.4439 g of barium sulphate. The percentage of sulphur in the compound is _____ (Nearest Integer)

(Given : Atomic mass Ba: 137 u, S: 32 u, O: 16 u)

Open solution

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Q19. Numerical

An athlete is given 100 g of glucose ($C_6H_{12}O_6$) for energy. This is equivalent to 1800kJ of energy. The 50% of this energy gained is utilized by the athlete for sports activities at the event. In order to avoid storage of energy, the weight of extra water he would need to perspire is _____ g (Nearest integer)

Assume that there is no other way of consuming stored energy.

Given : The enthalpy of evaporation of water is 45 kJ mol^{-1}

Molar mass of C, H & O are 12, 1 and 16 g mol^{-1}.

Open solution

Q20. Numerical

How many of the following metal ions have similar value of spin only magnetic moment in gaseous state? _____

(Given : Atomic number V, 23; Cr, 24; Fe, 26; Ni, 28)

V^{3+}, Cr^{3+}, Fe^{2+}, Ni^{3+}

Open solution

Q21. Numerical

The number of paramagnetic species from the following is _____.

$[Ni(CN)_4]^{2-}$, $[Ni(CO)_4]$, $[NiCl_4]^{2-}$

$[Fe(CN)_6]^{4-}$, $[Cu(NH_3)_4]^{2+}$

$[Fe(CN)_6]^{3-}$ and $[Fe(H_2O)_6]^{2+}$

Open solution

Q22. Numerical

For the first order reaction A $\rightarrow$ B, the half life is 30 min. The time taken for 75% completion of the reaction is _____ min. (Nearest integer)

Given : log 2 = 0.3010

log 3 = 0.4771

log 5 = 0.6989

Open solution

Q23. Numerical

The total number of lone pairs of electrons on oxygen atoms of ozone is _____.

Open solution

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Mathematics 23 questions

Q24. MCQ (Single Correct Answer)

The vector $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$ is rotated through a right angle, passing through the y-axis in its way and the resulting vector is $\vec{b}$. Then the projection of $3\vec{a} + \sqrt{2}\vec{b}$ on $\vec{c} = 5\hat{i} + 4\hat{j} + 3\hat{k}$ is :

(A) $\sqrt{6}$ (B) $2\sqrt{3}$ (C) 1 (D) $3\sqrt{2}$

Open solution

Q25. MCQ (Single Correct Answer)

The minimum value of the function $f(x) = \int_0^2 e^{|x-t|}dt$ is :

(A) 2 (B) $2(e-1)$ (C) $e(e-1)$ (D) $2e-1$

Open solution

Q26. MCQ (Single Correct Answer)

Let $x = 2$ be a local minima of the function $f(x) = 2x^4 - 18x^2 + 8x + 12, x \in (-4, 4)$. If M is local maximum value of the function f in $(-4, 4)$, then M =

(A) $18\sqrt{6} - \frac{33}{2}$ (B) $18\sqrt{6} - \frac{31}{2}$ (C) $12\sqrt{6} - \frac{33}{2}$ (D) $12\sqrt{6} - \frac{31}{2}$

Open solution

Q27. MCQ (Single Correct Answer)

The mean and variance of the marks obtained by the students in a test are 10 and 4 respectively. Later, the marks of one of the students is increased from 8 to 12. If the new mean of the marks is 10.2, then their new variance is equal to :

(A) 3.92 (B) 4.08 (C) 3.96 (D) 4.04

Open solution

Q28. MCQ (Single Correct Answer)

The value of $\lim\limits_{n\to\infty} \frac{1+2-3+4+5-6+.....+(3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$ is :

(A) $\frac{3}{2\sqrt{2}}$ (B) $\frac{3}{2}(\sqrt{2}+1)$ (C) $3(\sqrt{2}+1)$ (D) $\frac{\sqrt{2}+1}{2}$

Open solution

Q29. MCQ (Single Correct Answer)

Let M be the maximum value of the product of two positive integers when their sum is 66. Let the sample space $S = \{x \in \mathbb{Z} : x(66-x) \geq \frac{5}{9}M\}$ and the event $A = \{x \in S : x \text{ is a multiple of } 3\}$. Then P(A) is equal to :

(A) $\frac{1}{3}$ (B) $\frac{1}{5}$ (C) $\frac{7}{22}$ (D) $\frac{15}{44}$

Open solution

Q30. MCQ (Single Correct Answer)

Let $f(x) = \int \frac{2x}{(x^2+1)(x^2+3)}dx$. If $f(3) = \frac{1}{2}(\log_e 5 - \log_e 6)$, then $f(4)$ is equal to

(A) $\log_e 19 - \log_e 20$ (B) $\log_e 17 - \log_e 18$

(C) $\frac{1}{2}(\log_e 19 - \log_e 17)$ (D) $\frac{1}{2}(\log_e 17 - \log_e 19)$

Open solution

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Q31. MCQ (Single Correct Answer)

The distance of the point P(4, 6, -2) from the line passing through the point (-3, 2, 3) and parallel to a line with direction ratios 3, 3, -1 is equal to :

(A) 3 (B) $\sqrt{14}$ (C) $\sqrt{6}$ (D) $2\sqrt{3}$

Open solution

Q32. MCQ (Single Correct Answer)

Let $y(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$. Then $y' - y''$ at $x = -1$ is equal to

(A) 496 (B) 976 (C) 464 (D) 944

Open solution

Q33. MCQ (Single Correct Answer)

Consider the lines L_1 and L_2 given by

$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$

$L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$.

A line L_3 having direction ratios 1, -1, -2, intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is

(A) $4\sqrt{3}$ (B) $2\sqrt{6}$ (C) 4 (D) $3\sqrt{2}$

Open solution

Q34. MCQ (Single Correct Answer)

The points of intersection of the line $ax + by = 0, (a \neq b)$ and the circle $x^2 + y^2 - 2x = 0$ are $A(\alpha, 0)$ and $B(1, \beta)$. The image of the circle with AB as a diameter in the line $x + y + 2 = 0$ is :

(A) $x^2 + y^2 + 5x + 5y + 12 = 0$ (B) $x^2 + y^2 + 3x + 5y + 8 = 0$

(C) $x^2 + y^2 - 5x - 5y + 12 = 0$ (D) $x^2 + y^2 + 3x + 3y + 4 = 0$

Open solution

Q35. MCQ (Single Correct Answer)

Let $z_1 = 2 + 3i$ and $z_2 = 3 + 4i$. The set $S = \left\{z \in \mathbb{C} : |z - z_1|^2 - |z - z_2|^2 = |z_1 - z_2|^2\right\}$ represents a

(A) hyperbola with the length of the transverse axis 7

(B) hyperbola with eccentricity 2

(C) straight line with the sum of its intercepts on the coordinate axes equals -18

(D) straight line with the sum of its intercepts on the coordinate axes equals 14

Open solution

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Q36. MCQ (Single Correct Answer)

Open solution

Let S_1 and S_2 be respectively the sets of all $a \in \mathbb{R} - \{0\}$ for which the system of linear equations

$ax + 2ay - 3az = 1$

$(2a+1)x + (2a+3)y + (a+1)z = 2$

$(3a+5)x + (a+5)y + (a+2)z = 3$

has unique solution and infinitely many solutions. Then

(A) $n(S_1) = 2$ and S_2 is an infinite set

(B) $S_1 = \Phi$ and $S_2 = \mathbb{R} - \{0\}$

(C) $S_1 = \mathbb{R} - \{0\}$ and $S_2 = \Phi$

(D) S_1 is an infinite set and $n(S_2) = 2$

Q37. MCQ (Single Correct Answer)

Open solution

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} = \frac{y}{x}\left(1 + xy^2(1 + \log_e x)\right), x > 0, y(1) = 3$. Then $\frac{y^2(x)}{9}$ is equal to :

(A) $\frac{x^2}{5-2x^3(2+\log_e x^3)}$

(B) $\frac{x^2}{3x^3(1+\log_e x^2)-2}$

(C) $\frac{x^2}{7-3x^3(2+\log_e x^2)}$

(D) $\frac{x^2}{2x^3(2+\log_e x^3)-3}$

Q38. MCQ (Single Correct Answer)

Open solution

Let $f : (0, 1) \to \mathbb{R}$ be a function defined $f(x) = \frac{1}{1-e^{-x}}$, and $g(x) = (f(-x) - f(x))$. Consider two statements

(I) g is an increasing function in (0, 1)

(II) g is one-one in (0, 1)

Then,

(A) Both (I) and (II) are true

(B) Neither (I) nor (II) is true

(C) Only (II) is true

(D) Only (I) is true

Q39. Numerical

Open solution

The constant term in the expansion of $\left(2x + \frac{1}{x^7} + 3x^2\right)^5$ is _____.

Q40. Numerical

Open solution

Let S = {1, 2, 3, 5, 7, 10, 11}. The number of non-empty subsets of S that have the sum of all elements a multiple of 3, is _____.

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Q41. Numerical

Let x and y be distinct integers where $1 \leq x \leq 25$ and $1 \leq y \leq 25$. Then, the number of ways of choosing x and y, such that $x + y$ is divisible by 5, is _____.

Open solution

Q42. Numerical

For some a, b, c $\in \mathbb{N}$, let $f(x) = ax - 3$ and $g(x) = x^b + c, x \in \mathbb{R}$. If $(fog)^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$, then $(fog)(ac) + (gof)(b)$ is equal to _____.

Open solution

Q43. Numerical

Let $S = \left\{\alpha : \log_2(9^{2\alpha-4} + 13) - \log_2\left(\frac{5}{2} \cdot 3^{2\alpha-4} + 1\right) = 2\right\}$. Then the maximum value of β for which the equation $x^2 - 2\left(\sum_{\alpha \in S} \alpha\right)^2 x + \sum_{\alpha \in S} (\alpha + 1)^2 \beta = 0$ has real roots, is _____.

Open solution

Q44. Numerical

Let A_1, A_2, A_3 be the three A.P. with the same common difference d and having their first terms as $A, A + 1, A + 2$, respectively. Let a, b, c be the $7^{th}, 9^{th}, 17^{th}$ terms of A_1, A_2, A_3, respectively, such that $\begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0$.

If $a = 29$, then the sum of first 20 terms of an AP whose first term is $c - a - b$ and common difference is $\frac{d}{12}$, is equal to _____.

Open solution

Q45. Numerical

If the area enclosed by the parabolas $P_1 : 2y = 5x^2$ and $P_2 : x^2 - y + 6 = 0$ is equal to the area enclosed by P_1 and $y = \alpha x, \alpha > 0$, then α^3 is equal to _____.

Open solution

Q46. Numerical

If the sum of all the solutions of $\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \cot^{-1}\left(\frac{1-x^2}{2x}\right) = \frac{\pi}{3}, -1 < x < 1, x \neq 0$, is $\alpha - \frac{4}{\sqrt{3}}$, then α is equal to _____.

Open solution

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Physics 26 questions

Q47. MCQ (Single Correct Answer)

A parallel plate capacitor has plate area 40 cm^2 and plates separation 2 mm. The space between the plates is filled with a dielectric medium of a thickness 1 mm and dielectric constant 5. The capacitance of the system is :

Open solution

(A) $10\varepsilon_0$ F

(B) $24\varepsilon_0$ F

(C) $\frac{3}{10}\varepsilon_0$ F

(D) $\frac{10}{3}\varepsilon_0$ F

Q48. MCQ (Single Correct Answer)

Match List I with List II

Open solution

	List I		List II
A.	Surface tension	I.	$kg\,m^{-1}\,s^{-1}$
B.	Pressure	II.	$kg\,ms^{-1}$
C.	Viscosity	III.	$kg\,m^{-1}\,s^{-2}$
D.	Impulse	IV.	$kg\,s^{-2}$

Choose the correct answer from the options given below :

(A) A-IV, B-III, C-I, D-II

(B) A-IV, B-III, C-II, D-I

(C) A-II, B-I, C-III, D-IV

(D) A-III, B-IV, C-I, D-II

Q49. MCQ (Single Correct Answer)

The root mean square velocity of molecules of gas is

Open solution

(A) Proportional to temperature (T)

(B) Inversely proportional to square root of temperature $\left(\sqrt{\frac{1}{T}}\right)$

(C) Proportional to square of temperature (T^2)

(D) Proportional to square root of temperature ($\sqrt{T}$)

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Q50. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Photodiodes are used in forward bias usually for measuring the light intensity.

Reason R : For a p-n junction diode, at applied voltage V the current in the forward bias is more than the current in the reverse bias for $|V_z| > \pm V \geq |V_0|$ where V_0 is the threshold voltage and V_z is the breakdown voltage.

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true but R is NOT the correct explanation of A

(B) A is true but R is false

(C) Both A and R are true and R is the correct explanation of A

(D) A is false but R is true

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Q51. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List I (Current configuration)		List II (Magnitude of Magnetic Field at point O)
A.		I.	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 2]$
B.		II.	$B_0 = \frac{\mu_0}{4}\frac{I}{r}$
C.		III.	$B_0 = \frac{\mu_0 I}{2\pi r}[\pi - 1]$
D.		IV.	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 1]$

Choose the correct answer from the options given below :

(A) A-III, B-IV, C-I, D-II

(B) A-II, B-I, C-IV, D-III

(C) A-III, B-I, C-IV, D-II

(D) A-I, B-III, C-IV, D-II

Q52. MCQ (Single Correct Answer)

Open solution

In Young's double slits experiment, the position of 5th bright fringe from the central maximum is 5 cm. The distance between slits and screen is 1 m and wavelength of used monochromatic light is 600 nm. The separation between the slits is :

(A) 60 μm

(B) 48 μm

(C) 36 μm

(D) 12 μm

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Q53. MCQ (Single Correct Answer)

T is the time period of simple pendulum on the earth's surface. Its time period becomes x T when taken to a height R (equal to earth's radius) above the earth's surface. Then, the value of x will be :

(A) 4 (B) $\frac{1}{2}$ (C) 2 (D) $\frac{1}{4}$

Open solution

Q54. MCQ (Single Correct Answer)

Assume that the earth is a solid sphere of uniform density and a tunnel is dug along its diameter throughout the earth. It is found that when a particle is released in this tunnel, it executes a simple harmonic motion. The mass of the particle is 100 g. The time period of the motion of the particle will be (approximately)

(Take g = 10 m s^{-2} , radius of earth = 6400 km)

(A) 12 hours (B) 1 hour 24 minutes

(C) 24 hours (D) 1 hour 40 minutes

Open solution

Q55. MCQ (Single Correct Answer)

A car travels a distance of 'x' with speed v_1 and then same distance 'x' with speed v_2 in the same direction. The average speed of the car is :

(A) $\frac{v_1 v_2}{2(v_1+v_2)}$ (B) $\frac{2v_1 v_2}{v_1+v_2}$

(C) $\frac{2x}{v_1+v_2}$ (D) $\frac{v_1+v_2}{2}$

Open solution

Q56. MCQ (Single Correct Answer)

Electron beam used in an electron microscope, when accelerated by a voltage of 20 kV, has a de-Broglie wavelength of λ_0. IF the voltage is increased to 40 kV, then the de-Broglie wavelength associated with the electron beam would be :

(A) 3 λ_0 (B) 9 λ_0 (C) $\frac{\lambda_0}{\sqrt{2}}$ (D) $\frac{\lambda_0}{2}$

Open solution

Q57. MCQ (Single Correct Answer)

A uniform metallic wire carries a current 2 A, when 3.4 V battery is connected across it. The mass of uniform metallic wire is 8.92×10^{-3} kg, density is 8.92×10^3 kg/m^3 and resistivity is 1.7×10^{-8} Ω-m. The length of wire is :

(A) $l = 100$ m (B) $l = 6.8$ m (C) $l = 5$ m (D) $l = 10$ m

Open solution

Q58. MCQ (Single Correct Answer)

A solenoid of 1200 turns is wound uniformly in a single layer on a glass tube 2 m long and 0.2 m in diameter. The magnetic intensity at the center of the solenoid when a current of 2 A flows through it is :

(A) $1\ \mathrm{A\,m^{-1}}$ (B) $2.4 \times 10^{-3}\ \mathrm{A\,m^{-1}}$

(C) $1.2 \times 10^{3}\ \mathrm{A\,m^{-1}}$ (D) $2.4 \times 10^{3}\ \mathrm{A\,m^{-1}}$

Open solution

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Q59. MCQ (Single Correct Answer)

An electromagnetic wave is transporting energy in the negative z direction. At a certain point and certain time the direction of electric field of the wave is along positive y direction. What will be the direction of the magnetic field of the wave at that point and instant?

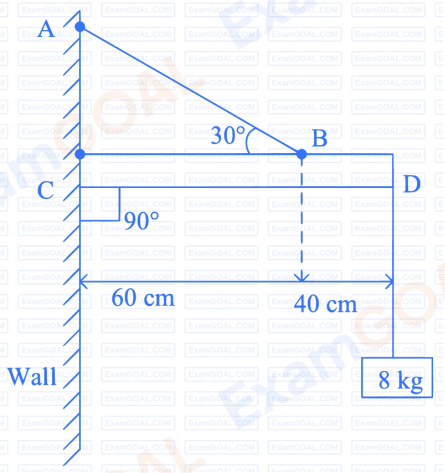
Open solution

(A) Negative direction of y
(B) Positive direction of z
(C) Positive direction of x
(D) Negative direction x

Q60. MCQ (Single Correct Answer)

An object of mass 8 kg is hanging from one end of a uniform rod CD of mass 2 kg and length 1 m pivoted at its end C on a vertical wall as shown in figure. It is supported by a cable AB such that the system is in equilibrium. The tension in the cable is (Take g = 10 m/s^2)

Open solution



(A) 90 N
(B) 240 N
(C) 30 N
(D) 300 N

Q61. MCQ (Single Correct Answer)

In an LC oscillator, if values of inductance and capacitance become twice and eight times, respectively, then the resonant frequency of oscillator becomes x times its initial resonant frequency ω_0. The value of x is :

Open solution

(A) 1/4
(B) 1/16
(C) 4
(D) 16

Q62. MCQ (Single Correct Answer)

A car is moving with a constant speed of 20 m/s in a circular horizontal track of radius 40 m. A bob is suspended from the roof of the car by a massless string. The angle made by the string with the vertical will be : (Take g = 10 m/s^2)

Open solution

(A) $\frac{\pi}{2}$
(B) $\frac{\pi}{6}$
(C) $\frac{\pi}{4}$
(D) $\frac{\pi}{3}$

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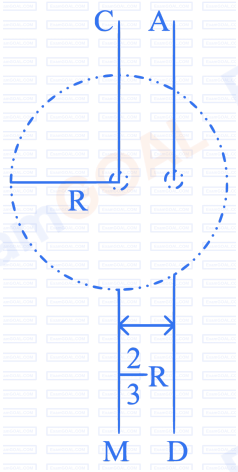
Q63. Numerical

If $\vec{P} = 3\hat{i} + \sqrt{3}\hat{j} + 2\hat{k}$ and $\vec{Q} = 4\hat{i} + \sqrt{3}\hat{j} + 2.5\hat{k}$ then, the unit vector in the direction of $\vec{P} \times \vec{Q}$ is $\frac{1}{x}\left(\sqrt{3}\hat{i} + \hat{j} - 2\sqrt{3}\hat{k}\right)$. The value of x is _____.

Open solution

Q64. Numerical

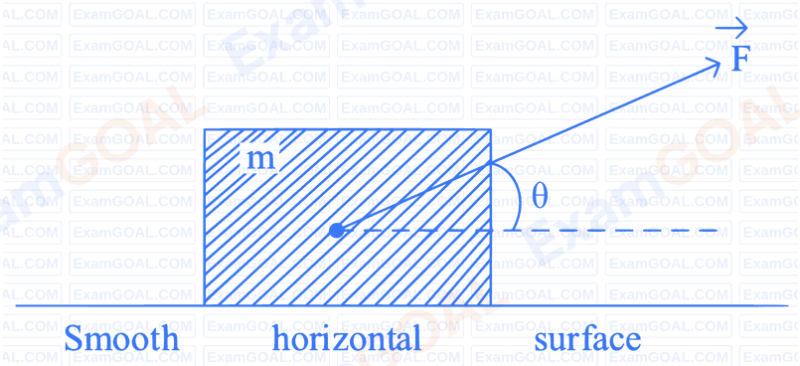
I_{CM} is the moment of inertia of a circular disc about an axis (CM) passing through its center and perpendicular to the plane of disc. I_{AB} is its moment of inertia about an axis AB perpendicular to plane and parallel to axis CM at a distance $\frac{2}{3}$R from center. Where R is the radius of the disc. The ratio of I_{AB} and I_{CM} is $x : 9$. The value of x is _____.



Open solution

Q65. Numerical

An object of mass 'm' initially at rest on a smooth horizontal plane starts moving under the action of force F = 2N. In the process of its linear motion, the angle θ (as shown in figure) between the direction of force and horizontal varies as $\theta = kx$, where k is a constant and x is the distance covered by the object from its initial position. The expression of kinetic energy of the object will be $E = \frac{n}{k}\sin\theta$. The value of n is _____.



Open solution

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Q66. Numerical

A ray of light is incident from air on a glass plate having thickness $\sqrt{3}$ cm and refractive index $\sqrt{2}$. The angle of incidence of a ray is equal to the critical angle for glass-air interface. The lateral displacement of the ray when it passes through the plate is _____ $\times 10^{-2}$ cm. (given $\sin 15^\circ = 0.26$)

Open solution

Q67. Numerical

The wavelength of the radiation emitted is λ_0 when an electron jumps from the second excited state to the first excited state of hydrogen atom. If the electron jumps from the third excited state to the second orbit of the hydrogen atom, the wavelength of the radiation emitted will be $\frac{20}{x}\lambda_0$. The value of x is _____.

Open solution

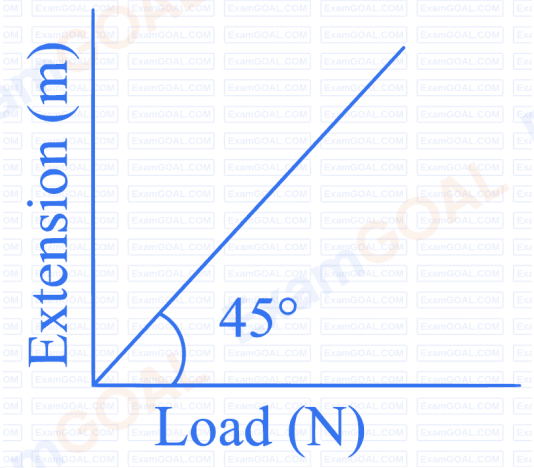
Q68. Numerical

The distance between two consecutive points with phase difference of 60° in a wave of frequency 500 Hz is 6.0 m. The velocity with which wave is travelling is _____ km/s

Open solution

Q69. Numerical

As shown in the figure, in an experiment to determine Young's modulus of a wire, the extension-load curve is plotted. The curve is a straight line passing through the origin and makes an angle of 45° with the load axis. The length of wire is 62.8 cm and its diameter is 4 mm. The Young's modulus is found to be $x \times 10^4$ Nm^{-2}. The value of x is _____.



Open solution

Q70. Numerical

An LCR series circuit of capacitance 62.5 nF and resistance of 50 Ω, is connected to an A.C. source of frequency 2.0 kHz. For maximum value of amplitude of current in circuit, the value of inductance is _____ mH.

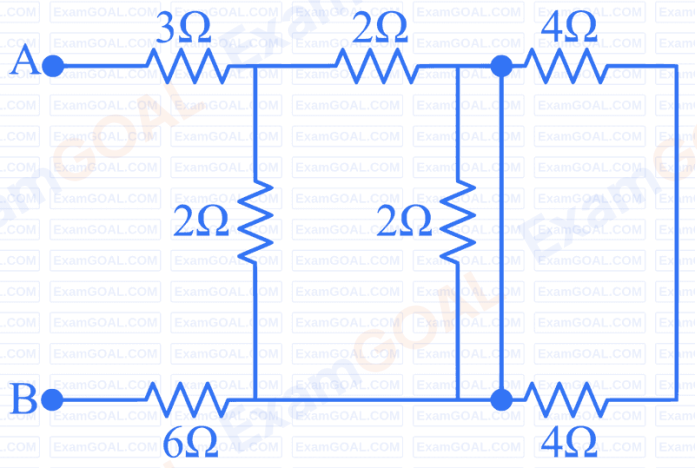
(Take $\pi^2 = 10$)

Open solution

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Q71. Numerical

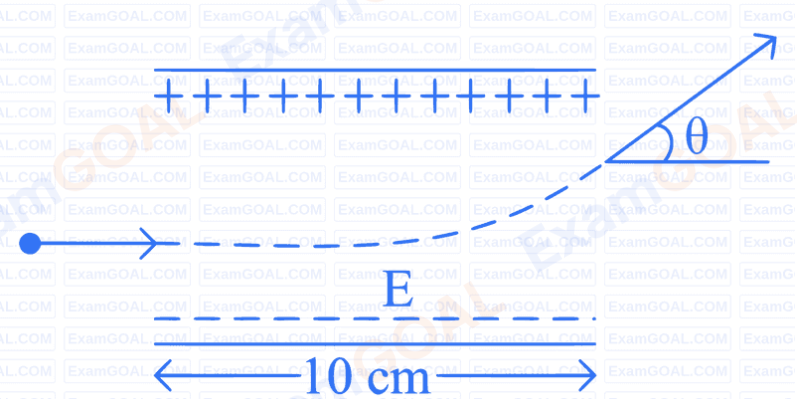
In the given circuit, the equivalent resistance between the terminal A and B is _____ Ω.



Open solution

Q72. Numerical

A uniform electric field of 10 N/C is created between two parallel charged plates (as shown in figure). An electron enters the field symmetrically between the plates with a kinetic energy 0.5 eV. The length of each plate is 10 cm. The angle (θ) of deviation of the path of electron as it comes out of the field is _____ (in degree).



Open solution

Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q73. MCQ (Single Correct Answer) · **Chemistry**

Reaction of thionyl chloride with white phosphorus forms a compound [A], which on hydrolysis gives [B], a dibasic acid. [A] and [B] are respectively :

(A) P_4O_6 and H_3PO_3 (B) $POCl_3$ and H_3PO_4 (C) PCl_5 and H_3PO_4 (D) PCl_3 and H_3PO_3

Open solution

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following reactions does not occur during extraction of copper?

(A) $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$

(B) $FeO + SiO_2 \rightarrow FeSiO_3$

(C) $CaO + SiO_2 \rightarrow CaSiO_3$

(D) $2Cu_2s + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Which of the following statements is incorrect for antibiotics?

(A) An antibiotic should promote the growth or survival of microorganisms.

(B) An antibiotic must be a product of metabolism.

(C) An antibiotic should be effective in low concentrations.

(D) An antibiotic is a synthetic substance produced as a structural analogue of naturally occurring antibiotic.

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Some reactions of NO_2 relevant to photochemical smog formation are



Identify A, B, X and Y :

(A) $X = [O], Y = NO, A = O_2, B = O_3$

(B) $X = N_2O, Y = [O], A = O_3, B = NO$

(C) $X = \frac{1}{2}O_2, Y = NO_2, A = O_3, B = O_2$

(D) $X = NO, Y = [O], A = O_2, B = N_2O_3$

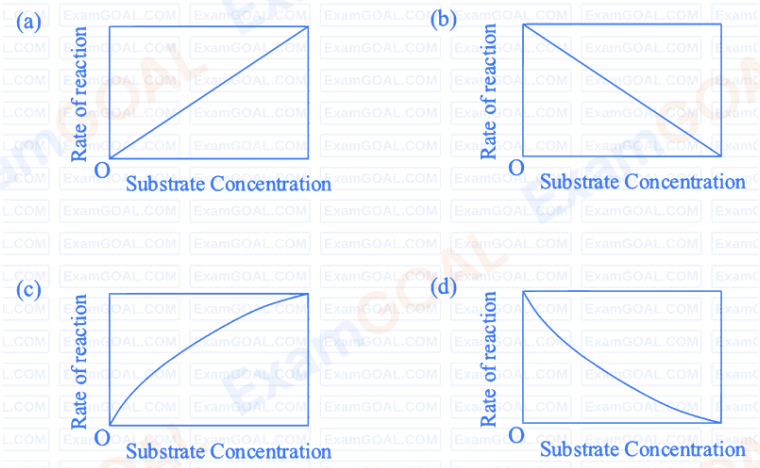
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Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The variation of the rate of an enzyme catalyzed reaction with substrate concentration is correctly represented by graph :



(A) (a) (B) (b) (C) (d) (D) (c)

Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Compound A reacts with NH_4Cl and forms a compound B. Compound B reacts with H_2O and excess of CO_2 to form compound C which on passing through or reaction with saturated NaCl solution forms sodium hydrogen carbonate. Compound A, B and C, are respectively :

(A) $Ca(OH)_2, NH_4^{\oplus}, (NH_4)_2CO_3$

(B) $CaCl_2, NH_4^{\oplus}, (NH_4)_2CO_3$

(C) $CaCl_2, NH_3, NH_4HCO_3$

(D) $Ca(OH)_2, NH_3, NH_4HCO_3$

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

A cubic solid is made up of two elements X and Y. Atoms of X are present on every alternate corner and one at the center of cube. Y is at $\frac{1}{3}^{rd}$ of the total faces. The empirical formula of the compound is :

(A) $X_2Y_{1.5}$ (B) X_3Y_2 (C) $XY_{2.5}$ (D) $X_{2.5}Y$

Q80. MCQ (Single Correct Answer) · **Mathematics**

Open solution

The statement $(p \wedge (\sim q)) \Rightarrow (p \Rightarrow (\sim q))$ is

(A) a tautology

(B) equivalent to $(\sim p) \vee (\sim q)$

(C) a contradiction

(D) $p \vee q$

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Q81. MCQ (Single Correct Answer) · **Mathematics**

If a_r is the coefficient of x^{10-r} in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^3\left(\frac{a_r}{a_{r-1}}\right)^2$ is equal to

(A) 3025 (B) 4895 (C) 5445 (D) 1210

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let $x, y, z > 1$ and $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$. Then $|\text{adj } (\text{adj } A^2)|$ is equal to

(A) 6^4 (B) 2^8 (C) 4^8 (D) 2^4

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non zero vectors such that $\vec{b} \cdot \vec{c} = 0$ and $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b}-\vec{c}}{2}$. If $\vec{d}$ be a vector such that $\vec{b} \cdot \vec{d} = \vec{a} \cdot \vec{b}$, then $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ is equal to

(A) $\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

The distance of the point $(6, -2\sqrt{2})$ from the common tangent $y = mx + c, m > 0$, of the curves $x = 2y^2$ and $x = 1 + y^2$ is :

(A) $\frac{1}{3}$ (B) 5 (C) $\frac{14}{3}$ (D) $5\sqrt{3}$

Open solution

Q85. Numerical · **Mathematics**

The vertices of a hyperbola H are $(\pm 6, 0)$ and its eccentricity is $\frac{\sqrt{5}}{2}$. Let N be the normal to H at a point in the first quadrant and parallel to the line $\sqrt{2}x + y = 2\sqrt{2}$. If d is the length of the line segment of N between H and the y-axis then d^2 is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

Let the equation of the plane passing through the line $x - 2y - z - 5 = 0 = x + y + 3z - 5$ and parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ be $ax + by + cz = 65$. Then the distance of the point (a, b, c) from the plane $2x + 2y - z + 16 = 0$ is _____.

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

A Carnot engine with efficiency 50% takes heat from a source at 600 K. In order to increase the efficiency to 70%, keeping the temperature of sink same, the new temperature of the source will be :

(A) 300 K (B) 1000 K (C) 900 K (D) 360 K

Open solution

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Q88. MCQ (Single Correct Answer) · **Physics**

The ratio of the density of oxygen nucleus $(^{16}_{8}O)$ and helium nucleus $(^{4}_{2}\mathrm{He})$ is

(A) 4 : 1 (B) 1 : 1 (C) 2 : 1 (D) 8 : 1

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

A message signal of frequency 5 kHz is used to modulate a carrier signal of frequency 2 MHz. The bandwidth for amplitude modulation is :

(A) 10 kHz (B) 2.5 kHz (C) 20 kHz (D) 5 kHz

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

A bowl filled with very hot soup cools from 98°C to 86°C in 2 minutes when the room temperature is 22°C. How long it will take to cool from 75°C to 69°C?

(A) 2 minutes (B) 0.5 minute (C) 1.4 minutes (D) 1 minute

Open solution

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Answer Key

Chemistry

1. **B**	2. **A**	3. **D**	4. **A**	5. **B**
6. **B**	7. **C**	8. **A**	9. **D**	10. **A**
11. **D**	12. **C**	13. **A**	14. **12**	15. **9079**
16. **10**	17. **41500**	18. **42**	19. **360**	20. **2**
21. **4**	22. **60**	23. **6**		

Mathematics

24. **D**	25. **B**	26. **C**	27. **C**	28. **B**
29. **A**	30. **D**	31. **B**	32. **A**	33. **B**
34. **A**	35. **D**	36. **C**	37. **A**	38. **A**
39. **1080**	40. **43**	41. **120**	42. **2039**	43. **25**
44. **495**	45. **600**	46. **2**		

Physics

47. **D**	48. **A**	49. **D**	50. **D**	51. **C**
52. **A**	53. **C**	54. **B**	55. **B**	56. **C**
57. **D**	58. **C**	59. **C**	60. **D**	61. **A**
62. **C**	63. **4**	64. **17**	65. **2**	66. **52**
67. **27**	68. **18**	69. **5**	70. **100**	71. **10**
72. **45**				

Out of syllabus

73. **D**	74. **C**	75. **A**	76. **A**	77. **D**
78. **D**	79. **B**	80. **A**	81. **D**	82. **B**
83. **C**	84. **B**	85. **216**	86. **9**	87. **B**
88. **B**	89. **A**	90. **C**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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